AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 21 · STUDENT LESSON

Tools and APIs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

State the contract. Validate the call. Check the result.

Inspect tool proposals without granting extra operations or inventing missing outputs.

Keep one habit: Room contract → book boundary → six simulated traces → bounded timeout policy → independent K cases.

MAIN LESSON · SECTION 1

What exactly may this tool do?

> Validate a proposed request before treating it as an executed operation.

Your learning goals

I can explain the **mechanism** and terminology of tools and **api contract**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall state and action scope

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Conversation State and Memory: a request needs current task state and source provenance. A previous selection is not permission for a new action; define the tool's operation and allowed arguments first.

Your end goal

By the end, I can explain tools and apis using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 21: Tools and APIs

A **tool call** is a structured request to an external capability with returned data.



CREATOR MISSION

Match tool schemas to requests and handle one failure.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Separate proposal, validation, execution and response

> A plausible call is not evidence that a tool ran.

Understand tools and APIs

A tool exposes an operation through a declared interface. An API is a way software sends requests and receives responses. A model-produced call is only a proposal: application checks decide whether the operation is allowed and its arguments valid. A successful read-only lookup supplies data; it does not authorise borrowing, booking or deleting. This lesson simulates requests on paper and makes no network calls.

Tools expose structured operations with **input** and **output** contracts. Generating a **tool call** is a proposal; validation and authorisation decide whether it executes.

Keep the exercise on supplied records

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use tools and apis to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Read the closed input and output contracts

> Zero, absent data and unavailable service are different outcomes.

The complete room-lookup contract

Toy lookup_room contract	Declared rule
Operation and permission	Read-only lookup of this public classroom registry; no booking or deletion
Request shape	Exactly one field room_id; nonempty string; no unknown fields
Registry	B has capacity 12; Z has measured capacity 0; all other room IDs absent
Successful result	found=true with matching room_id and exact integer capacity>=0
Missing result	found=false with matching room_id; no capacity measurement
Invalid request	Reject before invocation; do not silently remove extra operations
Transient timeout	At most2 total attempts including first, for this read-only operation only; then report unavailable
Do not retry	Invalid arguments, not found, denied access or invalid response; no invented success

Connect shape and action scope

For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts.

Keep missing information explicit

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

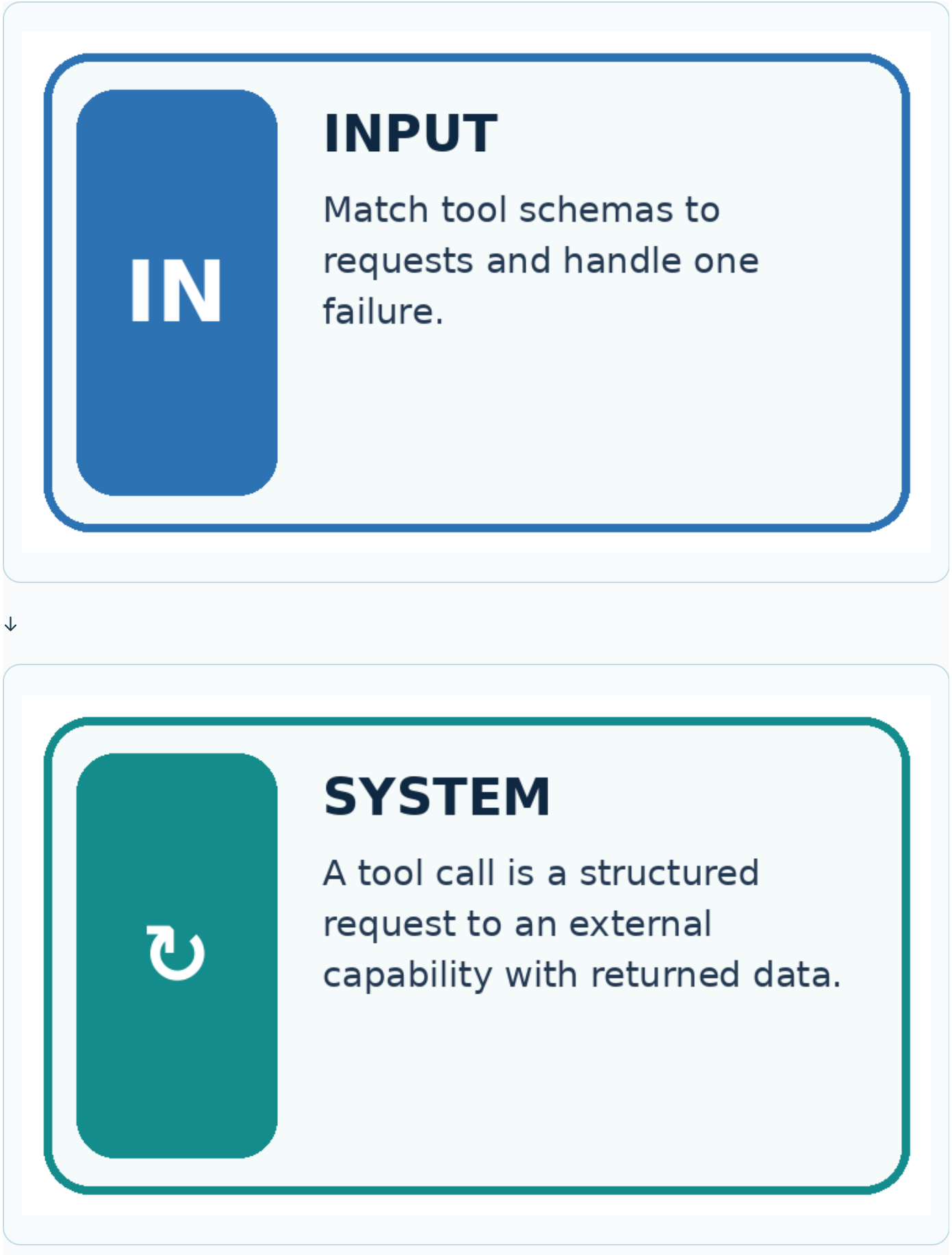
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

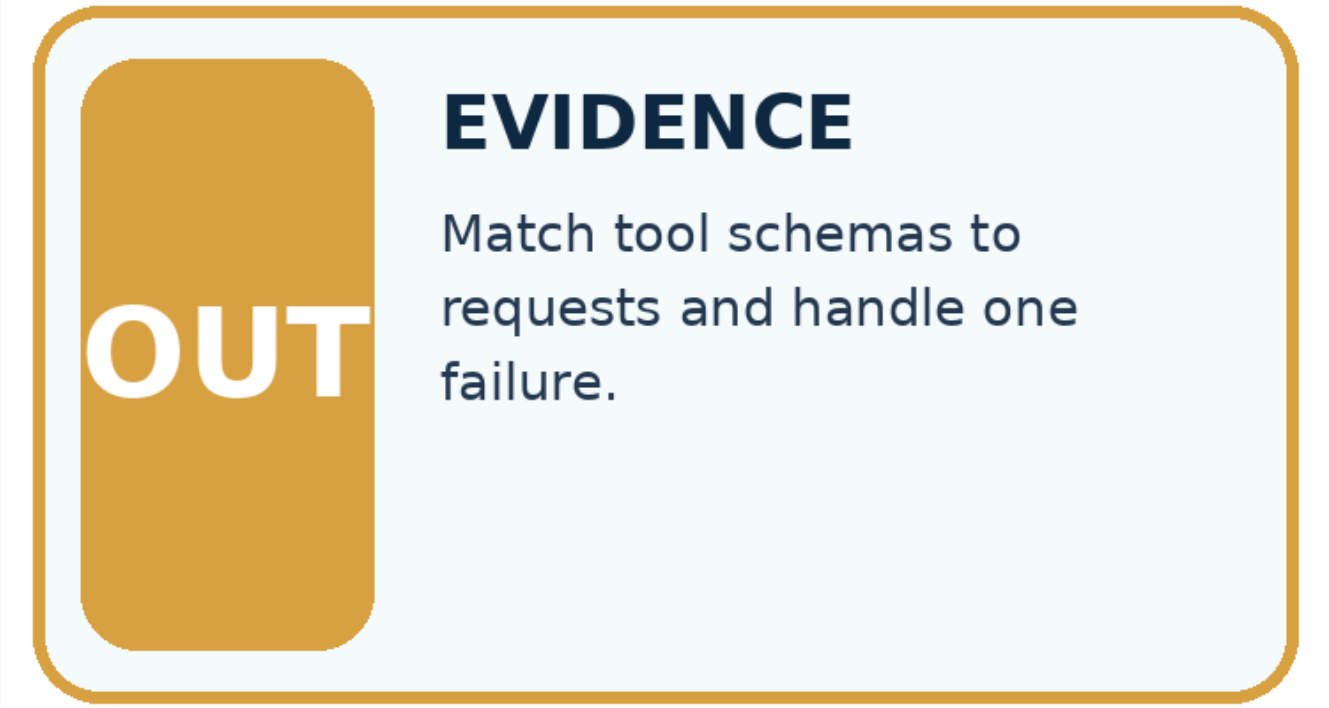
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Match tool schemas to requests and handle one failure.
↓
↻
SYSTEM
A tool call is a structured request to an external capability with returned data.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Tools and API contracts — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Proposed call lookup_room has room_id=B and extra argument book=true.	For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown arguments and do not replace missing values with invented facts.	Reject the extra field under the closed input contract. The lookup tool provides no booking authority even if the requested room exists.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

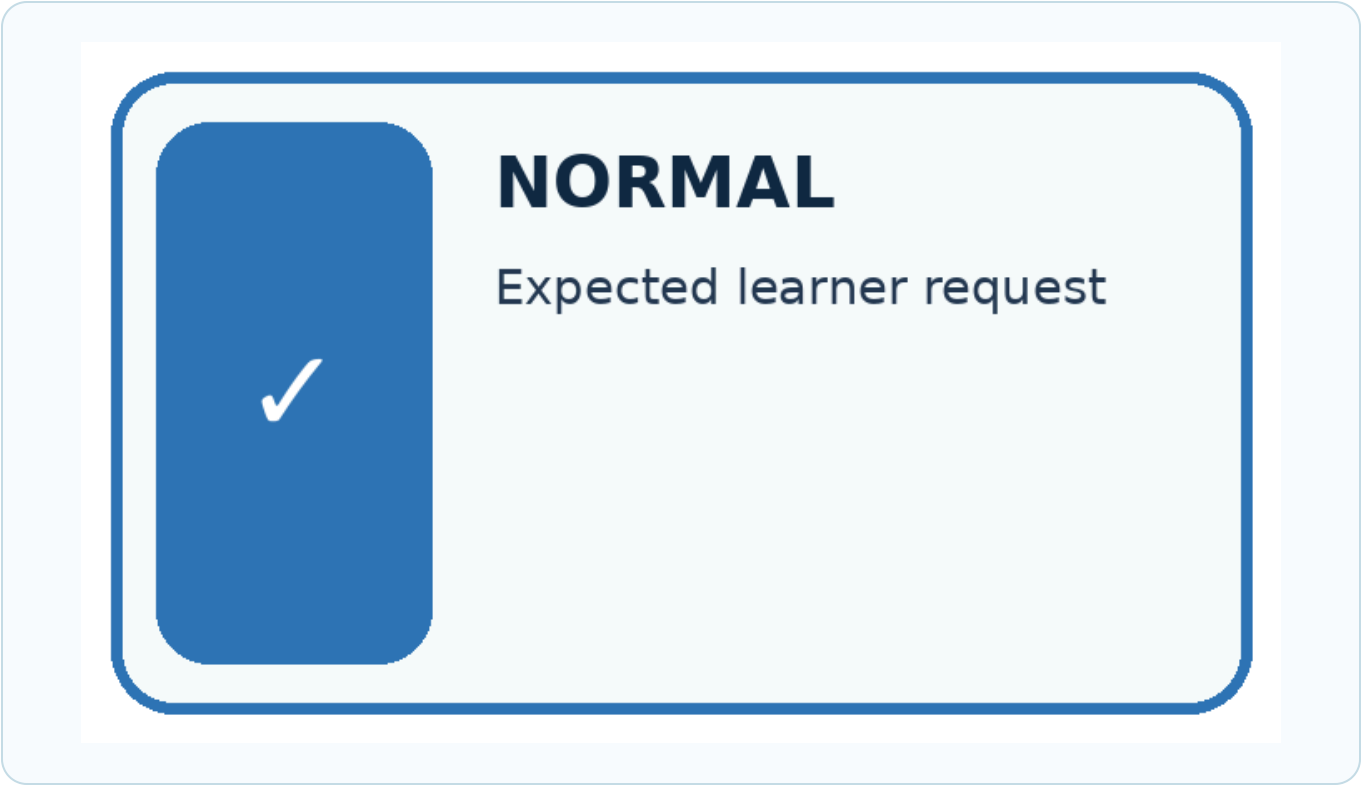
Connect the picture and the explanation

MAIN LESSON · SECTION 4

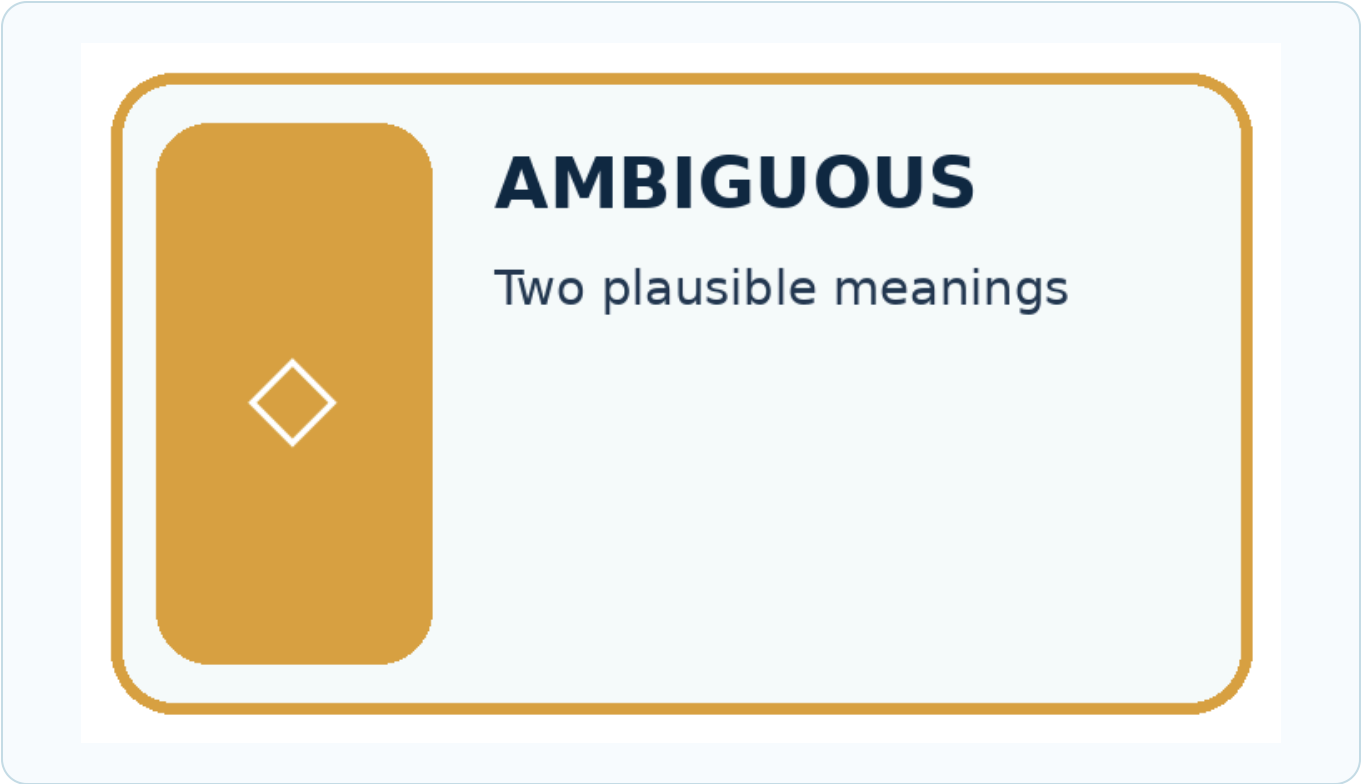
Compare valid input with supported operation

> A valid ID does not permit an extra action.

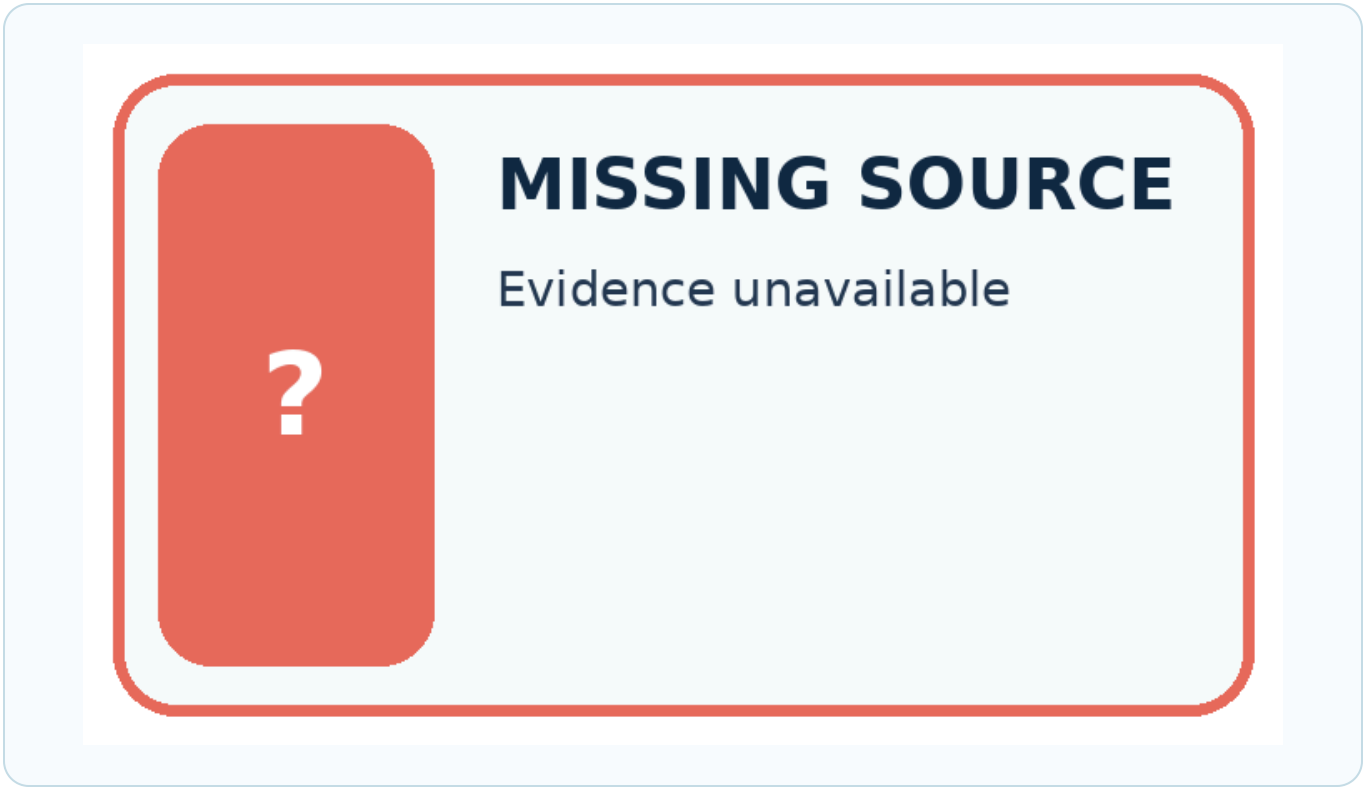
Four pictures with a purpose



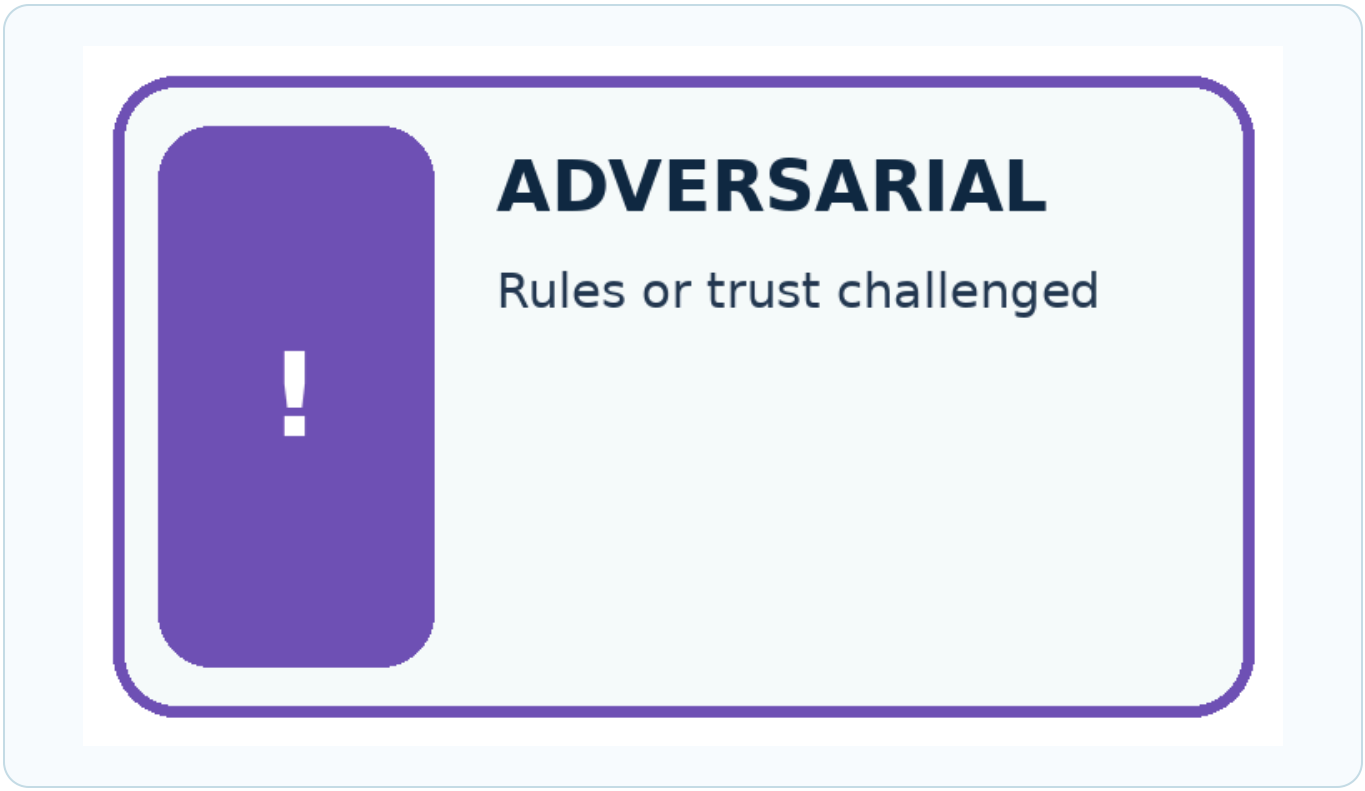
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Add lend: true to the same call.

▼ Check after predicting

Prediction: Reject the call before invoking the tool.

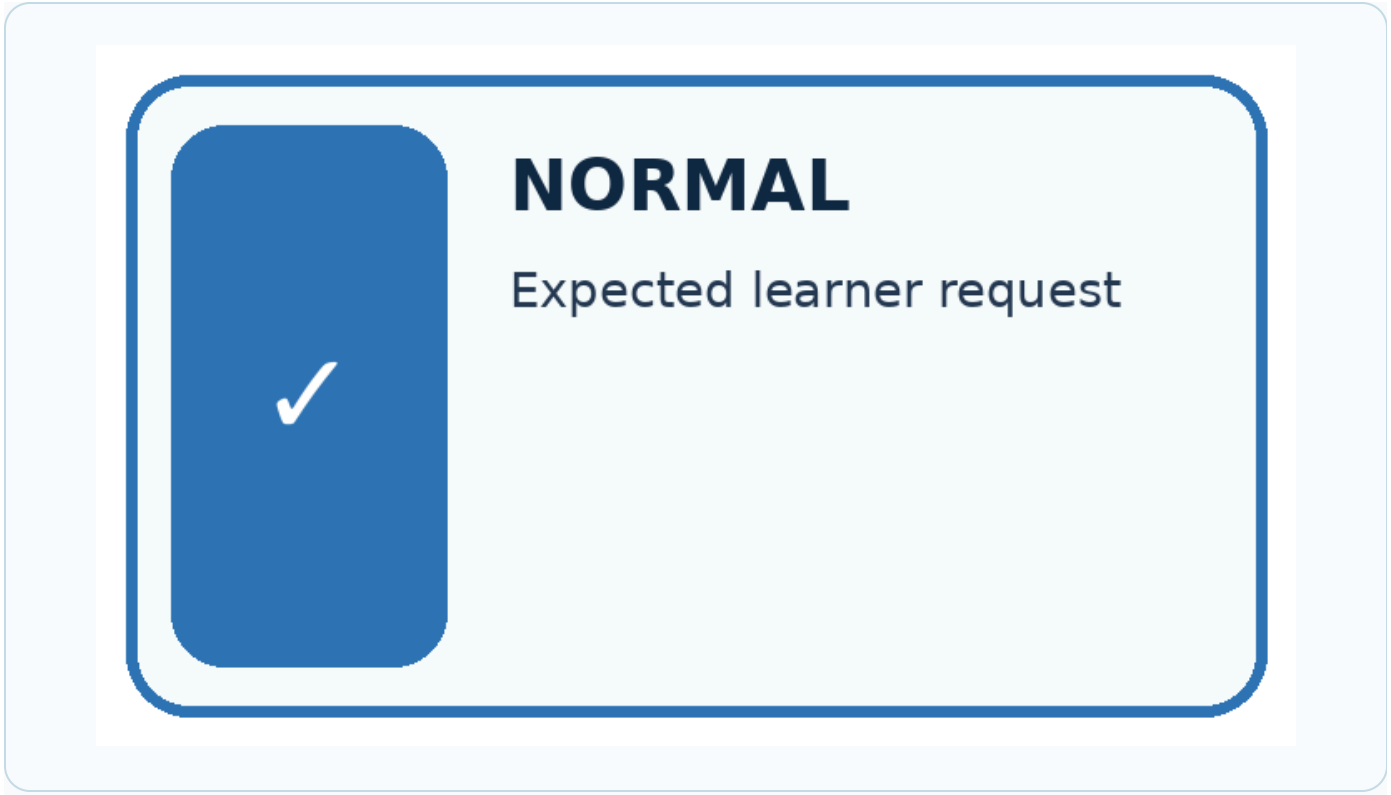
Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

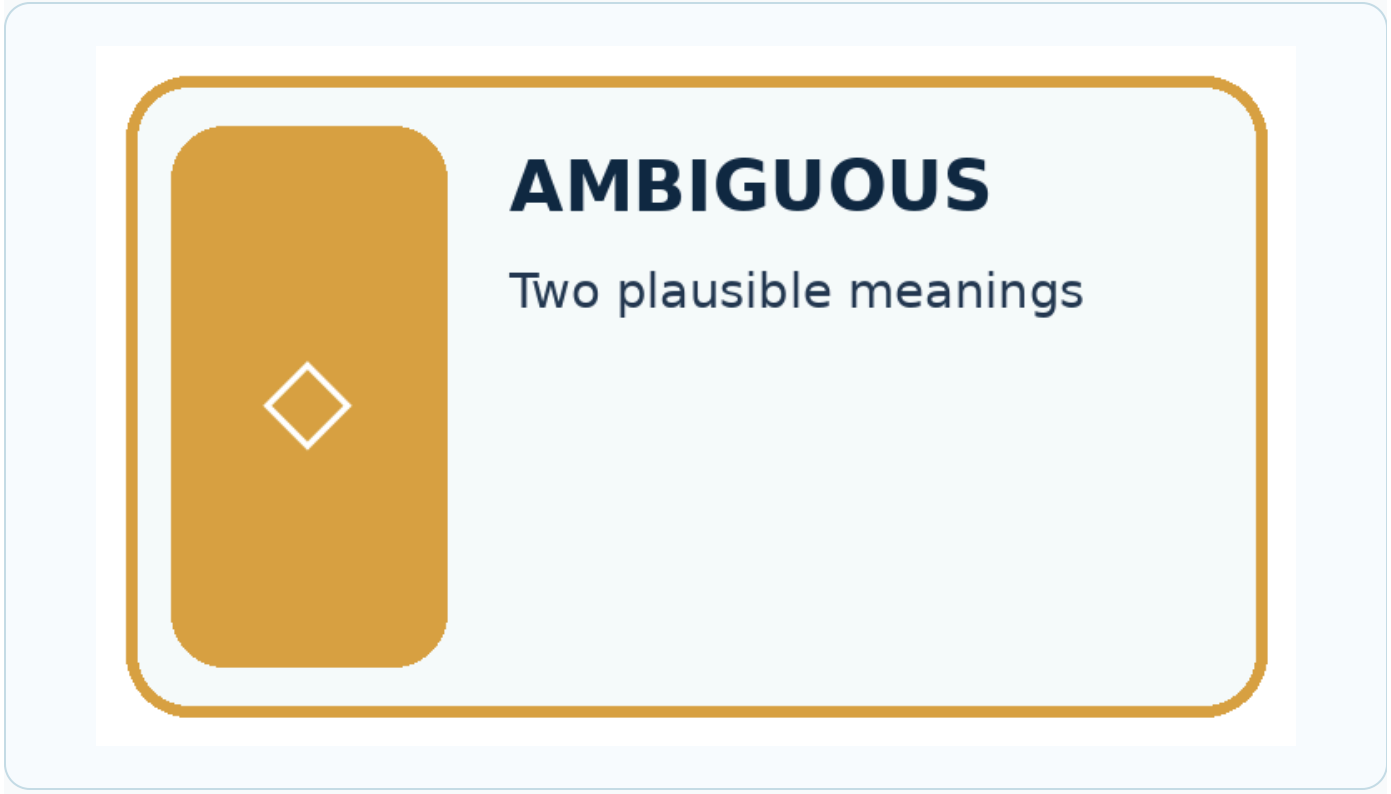
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the book request up to the operation boundary

> Do not invent a successful result from a valid request.

Read the book schema, IDs and scope

Check a library lookup call before it reaches a tool.

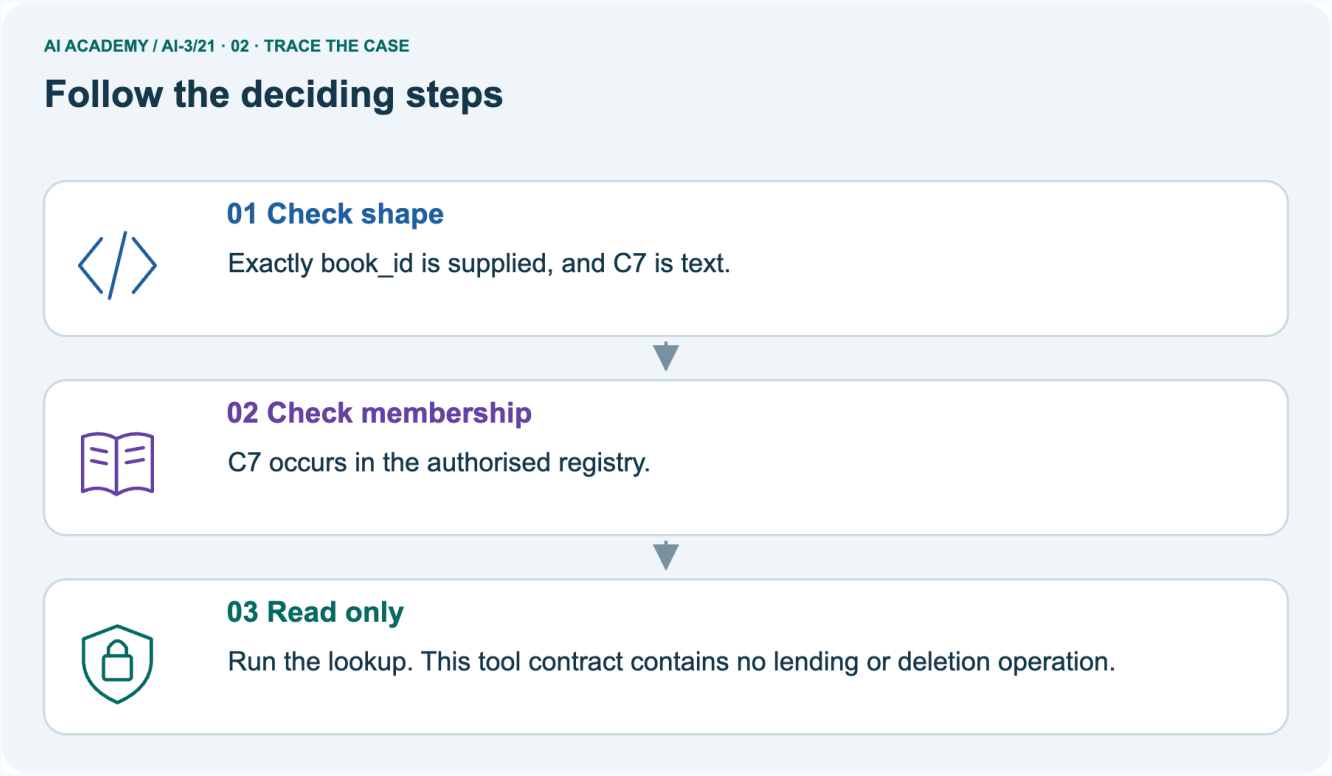
lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The separate lookup_book example is a membership-and-operation trace: supplied authorised registry IDs are C7 and M2, and the one allowed input field is book_id as text. The pictured call uses C7. No book-content record or actual response is supplied, so this trace establishes an eligible read-only request, not a completed lookup result. The room tool's capacity fields do not belong to the book schema.

Object notations in these diagrams are classroom shorthand; exact JSON syntax is introduced next week.

Trace shape, eligibility and read-only operation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check shape	Exactly book_id is supplied, and C7 is text.
2. Check membership	C7 occurs in the authorised registry.
3. Read only	Run the lookup. This tool contract contains no lending or deletion operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

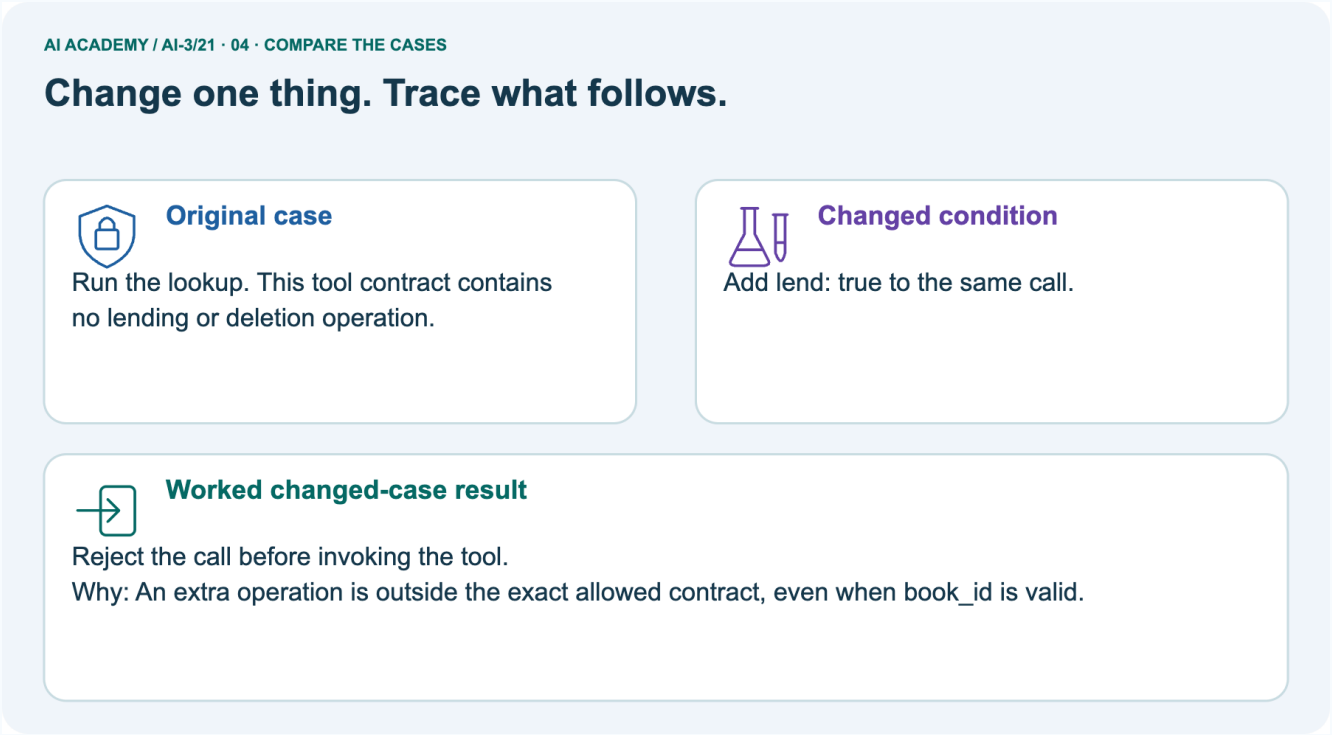
Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the extra lend field

- Change one thing: Add lend: true to the same call. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name response and access limits

- A valid **tool call** can still return unavailable or outdated records. Validate the response and access rights separately.

- Read the labelled picture: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Extra book argument to room lookup

Proposed call lookup_room has room_id=B and extra **argument** book=true.

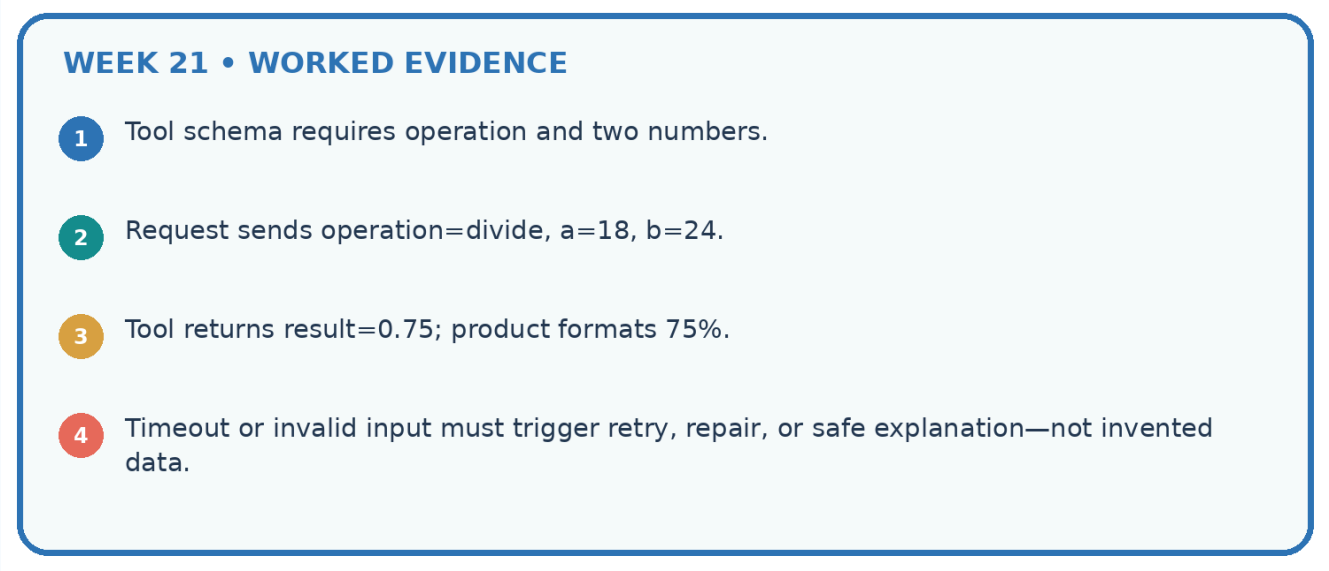
Reject the extra field under the closed **input** contract. The lookup tool provides no booking authority even if the requested room exists.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Tool schema requires operation and two numbers.

2. Request sends operation=divide, a=18, b=24.

3. Tool returns result=0.75; product formats 75%.

4. Timeout or invalid **input** must trigger retry, repair, or safe explanation—not invented data.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

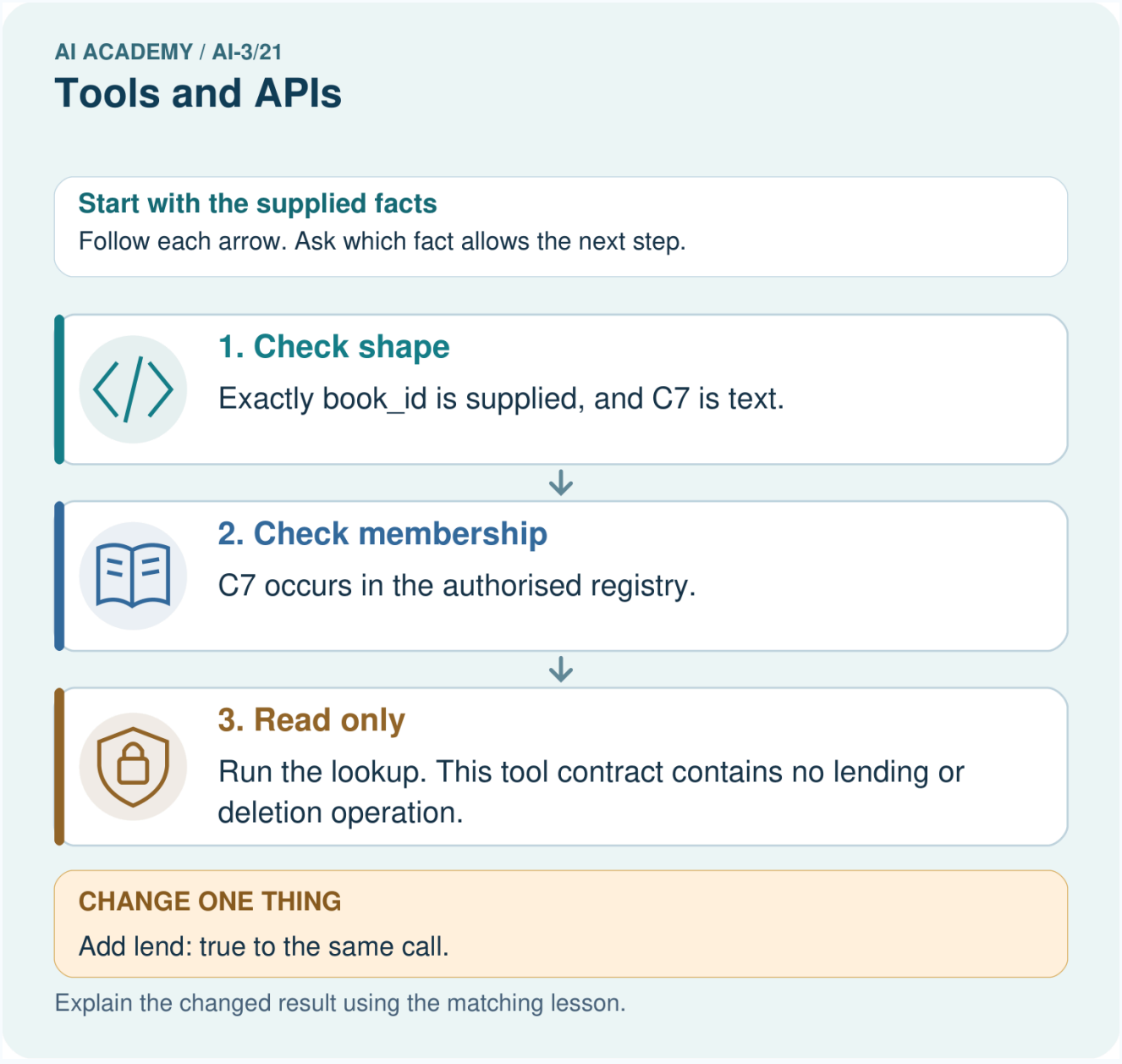
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Tools and APIs

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find an operation hidden in an argument

> Membership, schema validity and permission are separate checks.

Repair the extra-action and missing-value shortcuts

Claim to repair: The ID exists, so ignore the extra booking field and execute the request. The contract accepts exactly one input field. Reject the extra operation before invocation rather than silently rewriting its meaning. A read-only result provides data only. Another shortcut is treating not found as capacity zero: Z has a measured zero, whereas an absent room has no capacity value.

Validate response ID, status and fields before using returned data.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A model can use any tool just by describing the desired action. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise request validation with less help

> Use the closed room contract; keep book and room schemas separate.

The result-versus-authority limit

A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

Complete the extra-field W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Tools and **API contract**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit a guarantee shortcut

- Use the worked case for Tools and **API contract**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured action boundary

- Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
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Keep independent cases separate

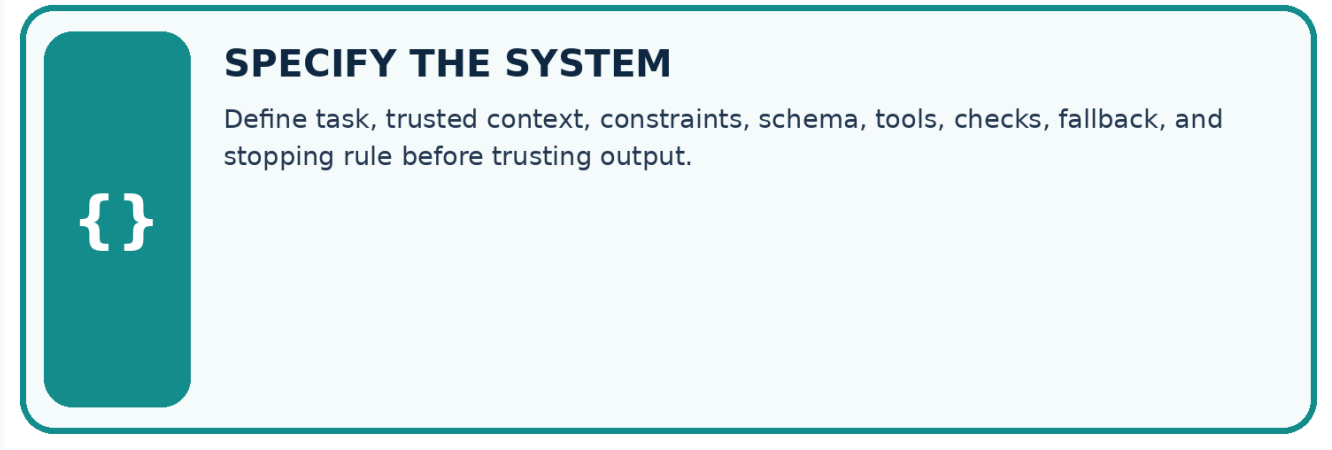
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, **tool call**s, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Simulate six complete tool traces

Keep request rejection, response rejection and unavailable service distinct.

All supplied request and response cards

Supplied fictional card	Expected trace under the room contract
A · Request room_id=B; returned found=true, room_id=B, capacity 12.	Valid request and response; report supplied capacity 12, no booking.
B · Request room_id=Z; returned found=true, room_id=Z, capacity 0.	Valid measured zero; do not treat 0 as missing.
C · Request room_id=B and book=true.	Reject extra field before invoking.
D · Request room_id as numeric12.	Reject wrong type before invoking.
E · Valid request B; attempt 1 timeout, attempt 2 timeout.	Two attempts exhausted; unavailable, no third attempt or invented data.

Supplied fictional card	Expected trace under the room contract
F · Valid request B; returned found=true, room_id=Z, capacity 0.	Reject mismatched response identity; do not use another room's data.

Predict, validate and trace

1. Cover the expected trace and classify each complete card. Record request validation, permitted operation, whether invocation is allowed, response check and final reported status. Uncover the supplied guidance afterwards. The timeout budget is two total attempts, not two retries after the first; it applies only to this read-only transient-timeout case. Do not extend that retry policy to side-effecting tools.

   Unknown IDs in the room contract are valid strings that produce not found, distinct from invalid input. No network calls, bookings or measured tool executions occur here. Preserve the independent K cases for the first uncoached attempt.

Keep the paper evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

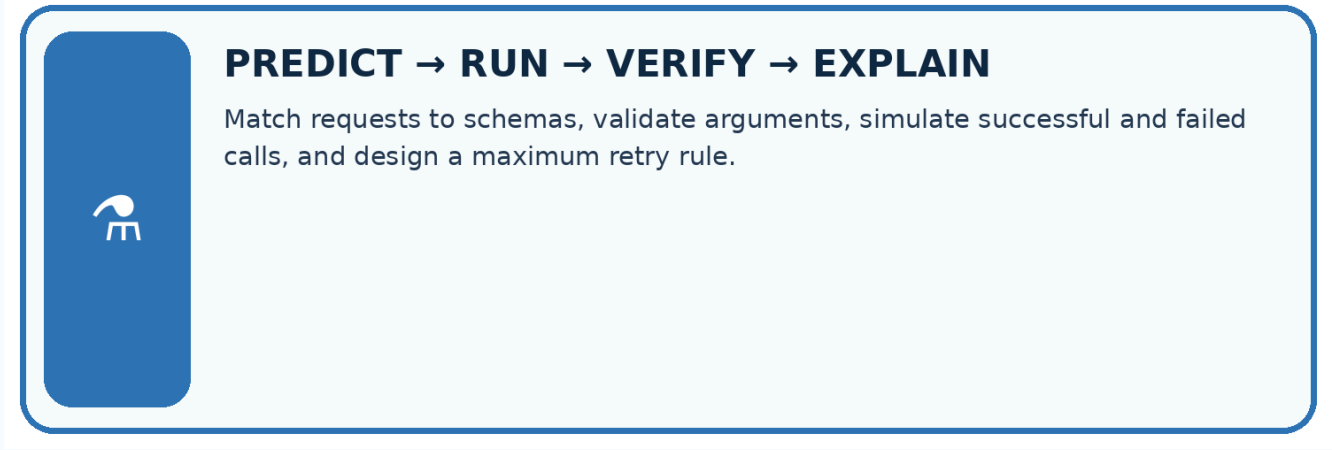
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the six supplied toy room-lookup cards and complete closed contract. Validate **argument**s, action scope and response identity; distinguish measured zero from missing and stop after at most two total read-only timeout attempts. Record paper traces, not live calls.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: distinguish type, missing record and extra action

> Use the supplied K cases under the declared room contract.

Independent cases and syntax scope

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

Apply the same room input contract: exactly room_id as a nonempty string; extra fields are rejected before invocation. Treat the K values in the task's shorthand as text. K is absent from the registry. No real API, deletion capability or capacity measurement for K is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Allow only retrieval and calculator tools with validated **argument**s and visible results.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A tool can calculate or search supplied notes. The request is 'What is 17×6, and where do roots take up water?' Assign each subtask and describe **output** checks.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's tool boundary inspectable

> Validate both directions before using a returned value.

This week's contribution

For the Verified Study Buddy, list each tool's operation, permission boundary, exact request/response fields, validation rules, failure statuses and bounded retry policy. Preserve the original request and returned ID so a response cannot silently attach to a different object. Keep lookup separate from actions that change records.

A generated call, eligible request and validated result are three different evidence stages.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Allow only retrieval and calculator tools with validated **argument**s and visible results.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

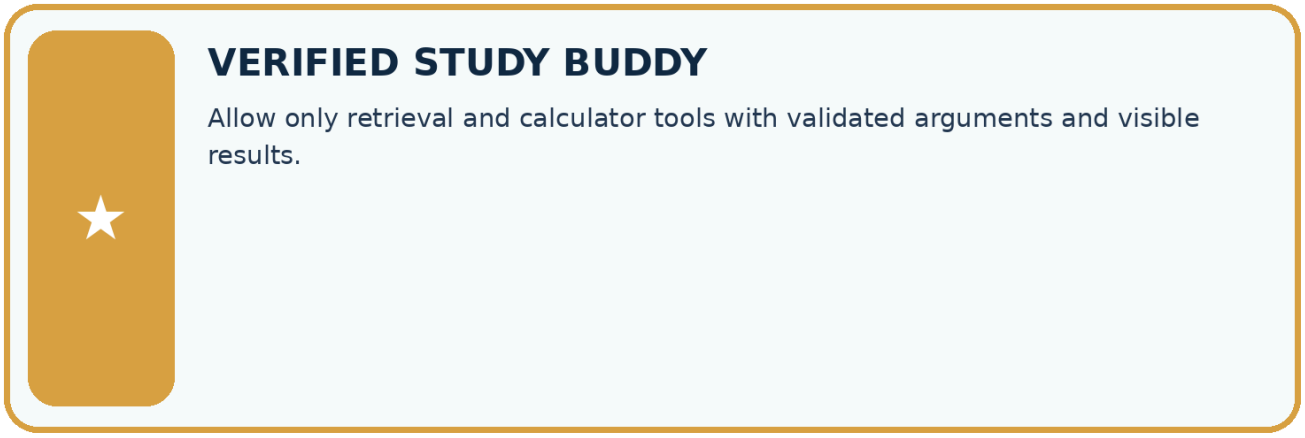
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: say what was proposed, allowed and established

> A valid request is not a validated response or a permission for another operation.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of tools and **api contract**s

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
API	An agreed interface through which software requests data or actions from another system.	
tool call	A structured request to a tool with specified arguments.	
argument	A value supplied to a function or tool to tell it what to operate on.	
error response	A message or structured result indicating that a request could not be completed as expected.	

Terms for the connected explanation

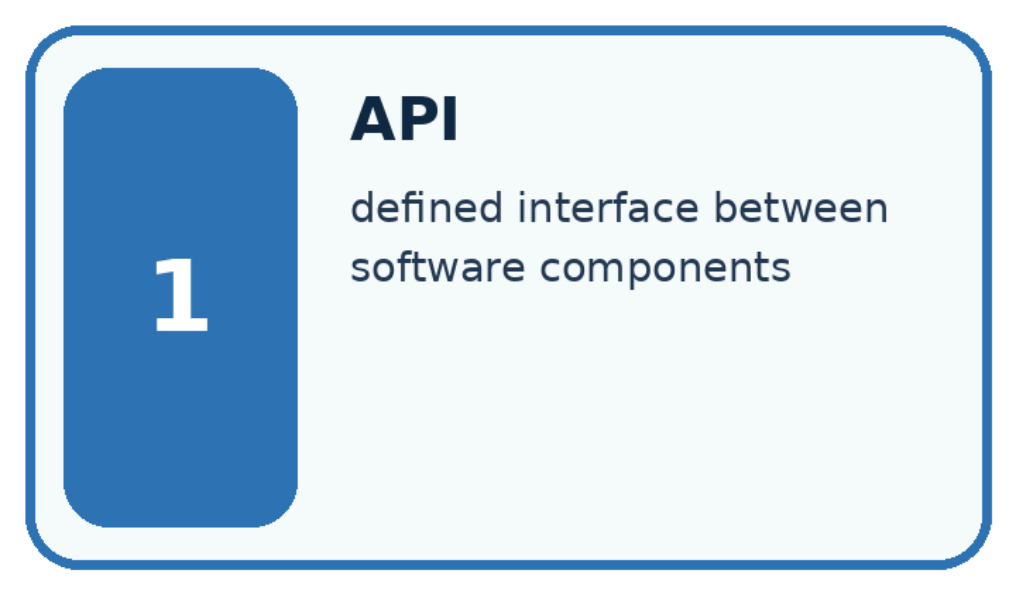
Word	Meaning
API contract	Declared input, output and failure rules for an interface
Not found	An explicit absence result rather than a fabricated value

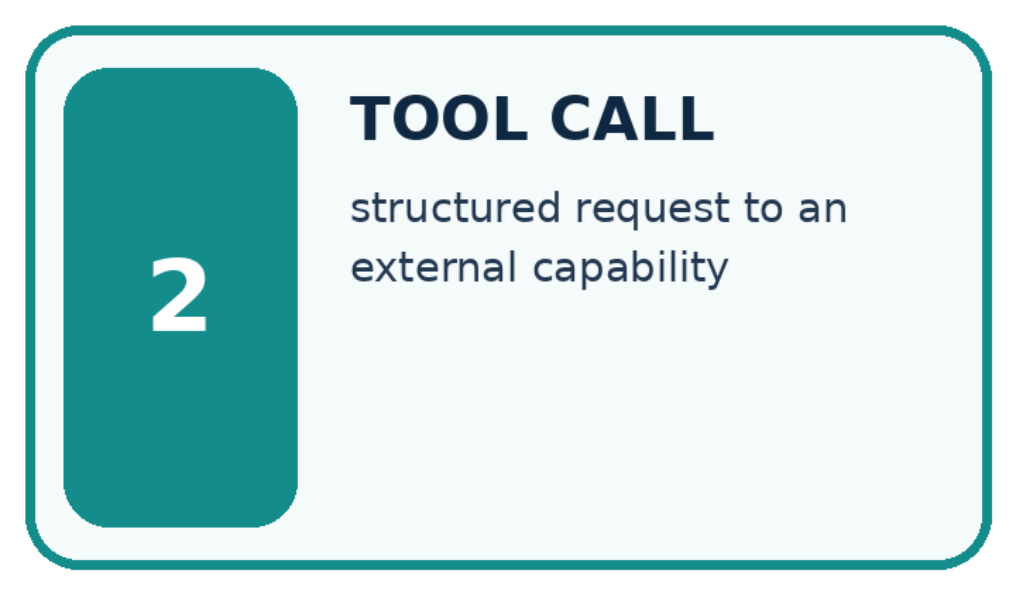
▼ **Earlier wording, alternatives and picture transcripts**

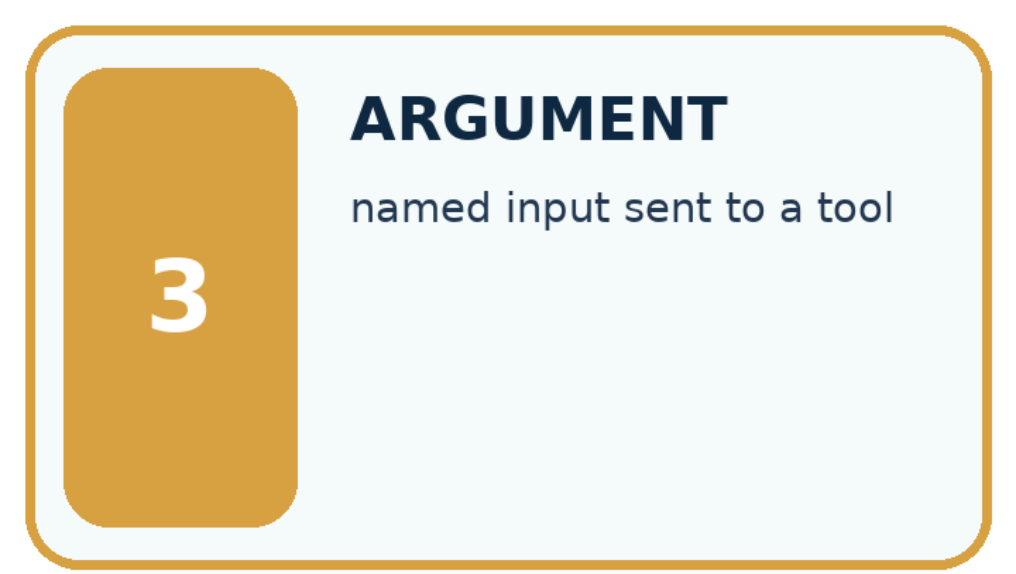
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.









Words used in the closer explanation

▼ Choose a stronger thinking challenge

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

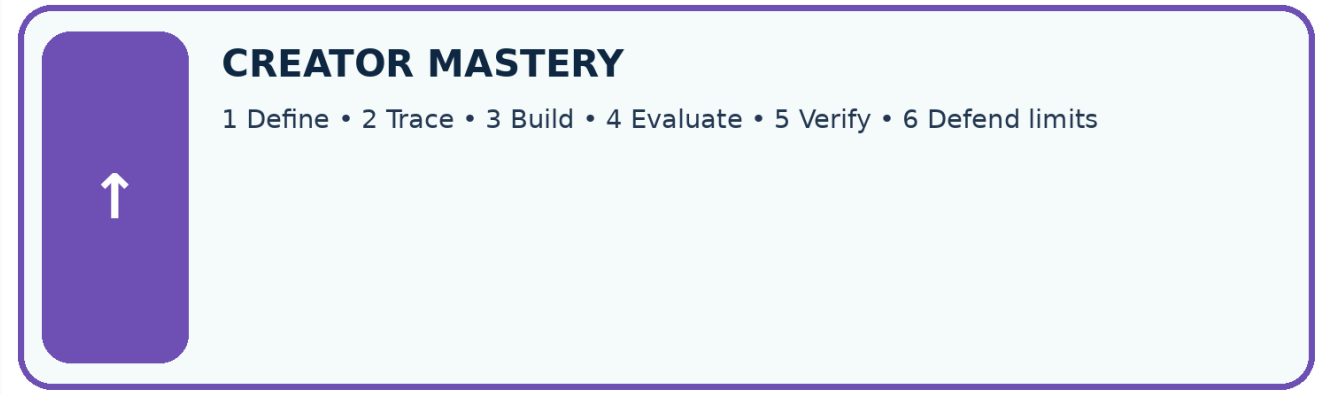
> **Write or draw your reasoning and evidence.**

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Validate the fresh K room calls independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check and record

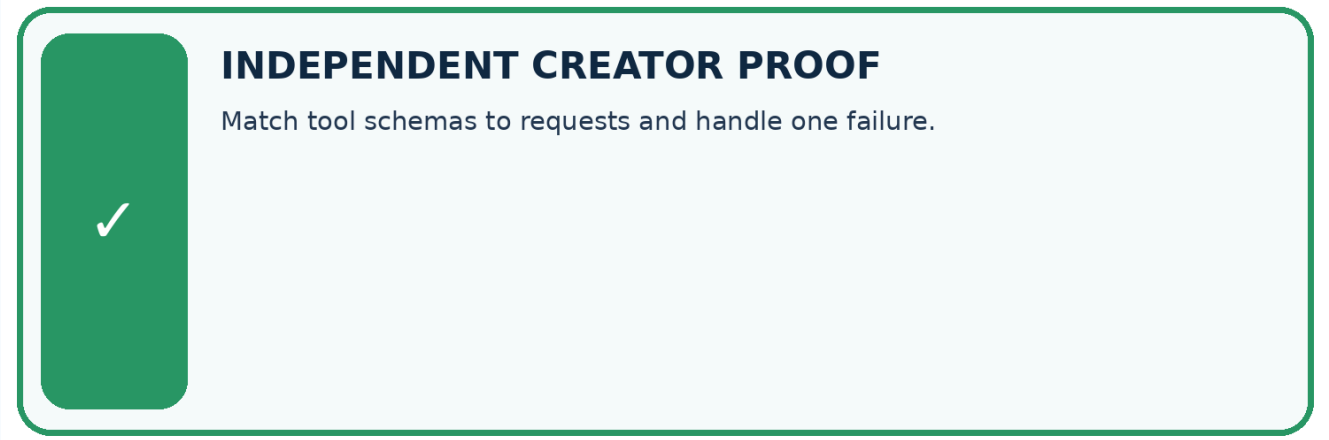
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week’s idea

Use after practice. Validate the fresh K room calls independently is the selected independent case.



EXPLANATION CHECK Require input type/field checks, not-found semantics and a separate action boundary.

> **Write or draw your reasoning and evidence.**

▼ **Extra practice: Choose a tool for each subtask**

REFERENCE · SECTION 13

Extra practice: Choose a tool for each subtask

> Try the question before opening its worked explanation.

Read the supplied case and question

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Check after trying

Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

Evidence or stage	Value or result
12 times 9	Calculator → 108
Habitat definition	Search approved class pack
Before answering	Verify input, result and support

Earlier wording, alternatives and picture transcripts

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose a tool for each subtask

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Calculator returns 108; search retrieves the definition. Verify **input**s, **output**s and supporting passage before answering.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of tools and api contracts	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Check a library lookup call before it reaches a tool.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Check a library lookup call before it reaches a tool.
- Which labelled step matches this detail? Run the lookup. This tool contract contains no lending or deletion operation.
- 1. Check shape
- 2. Read only
- 3. Check membership
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Read only Run the lookup. This tool contract contains no lending or deletion operation. Read the picture in order. Run the lookup.

  This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Add lend: true to the same call. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Reject the call before invoking the tool.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Conversation State and Memory (ai-3/20). Goal: Explain the **mechanism** and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

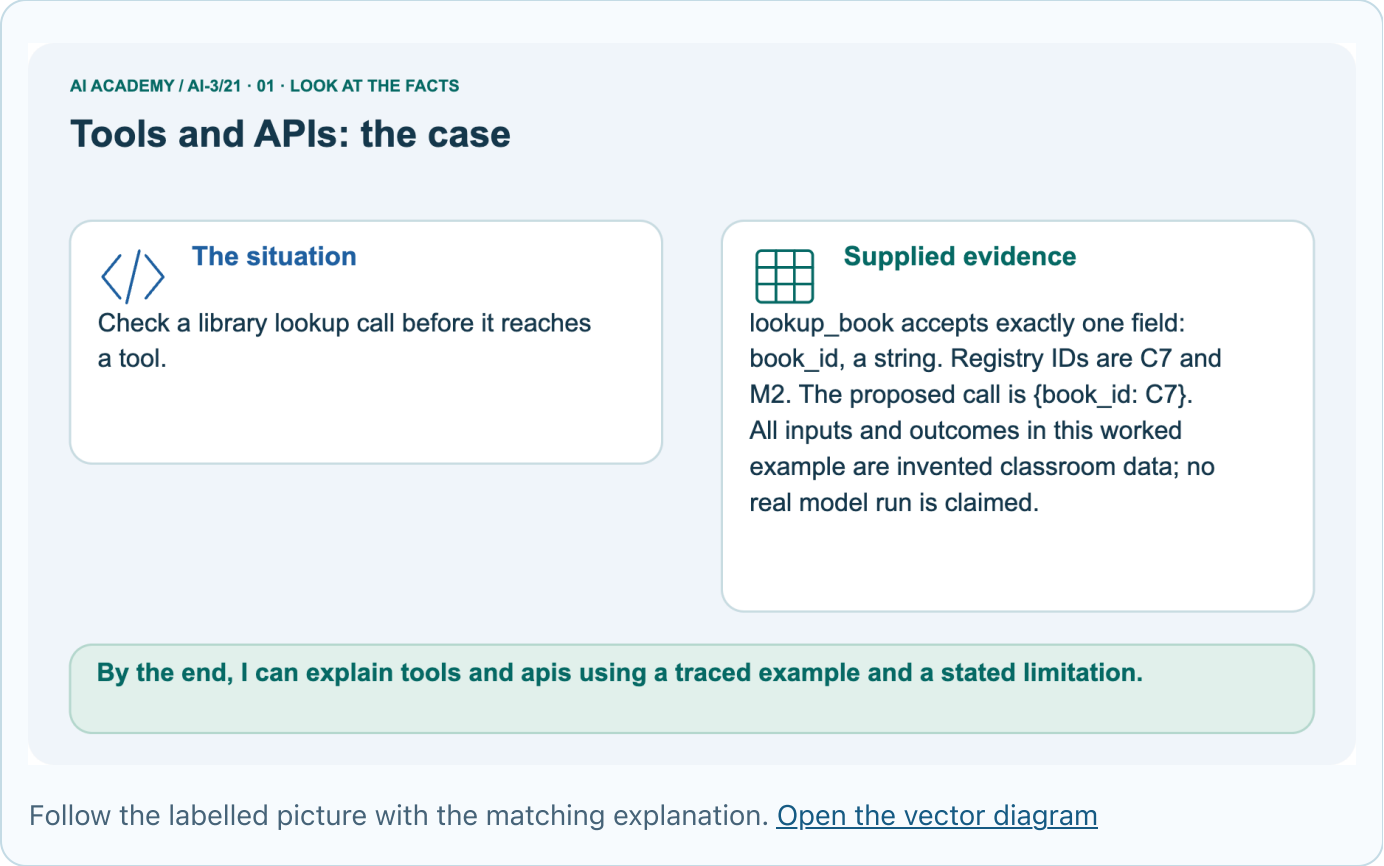
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Tools and APIs: the case



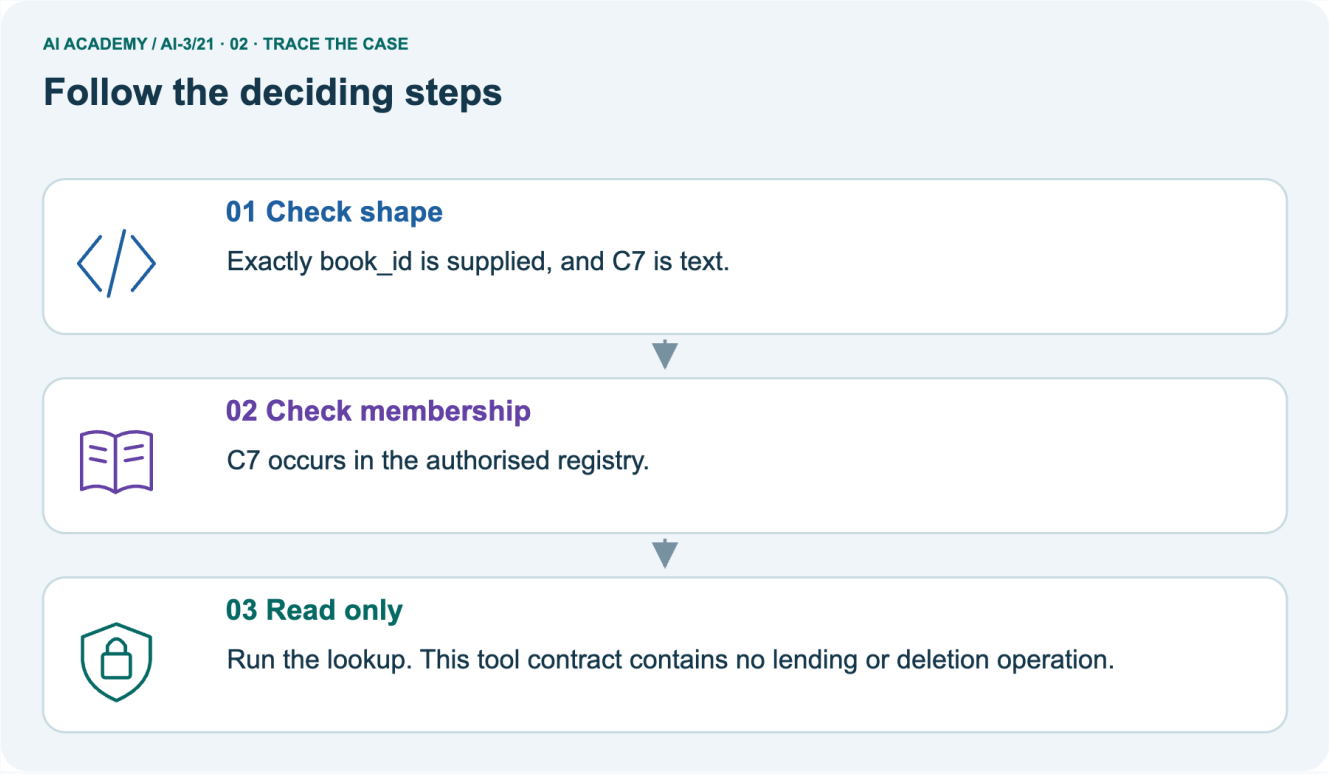
Follow the labelled picture with the matching explanation. Open the vector diagram

Check a library lookup call before it reaches a tool. Supplied facts: lookup_book accepts exactly one field: book_id, a string. Registry IDs are C7 and M2. The proposed call is {book_id: C7}.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

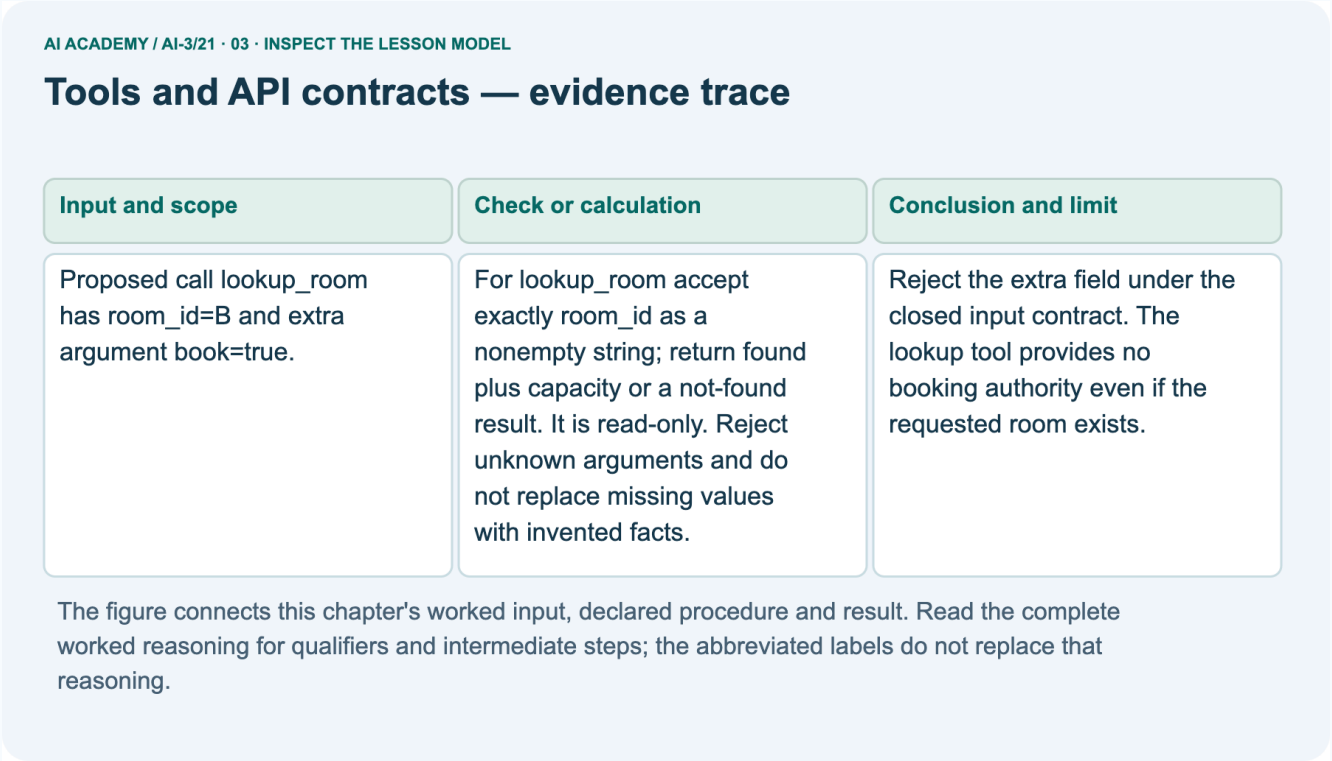


Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape: Exactly book_id is supplied, and C7 is text. → Check membership: C7 occurs in the authorised registry. → Read only: Run the lookup. This tool contract contains no lending or deletion operation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Tools and API contracts — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

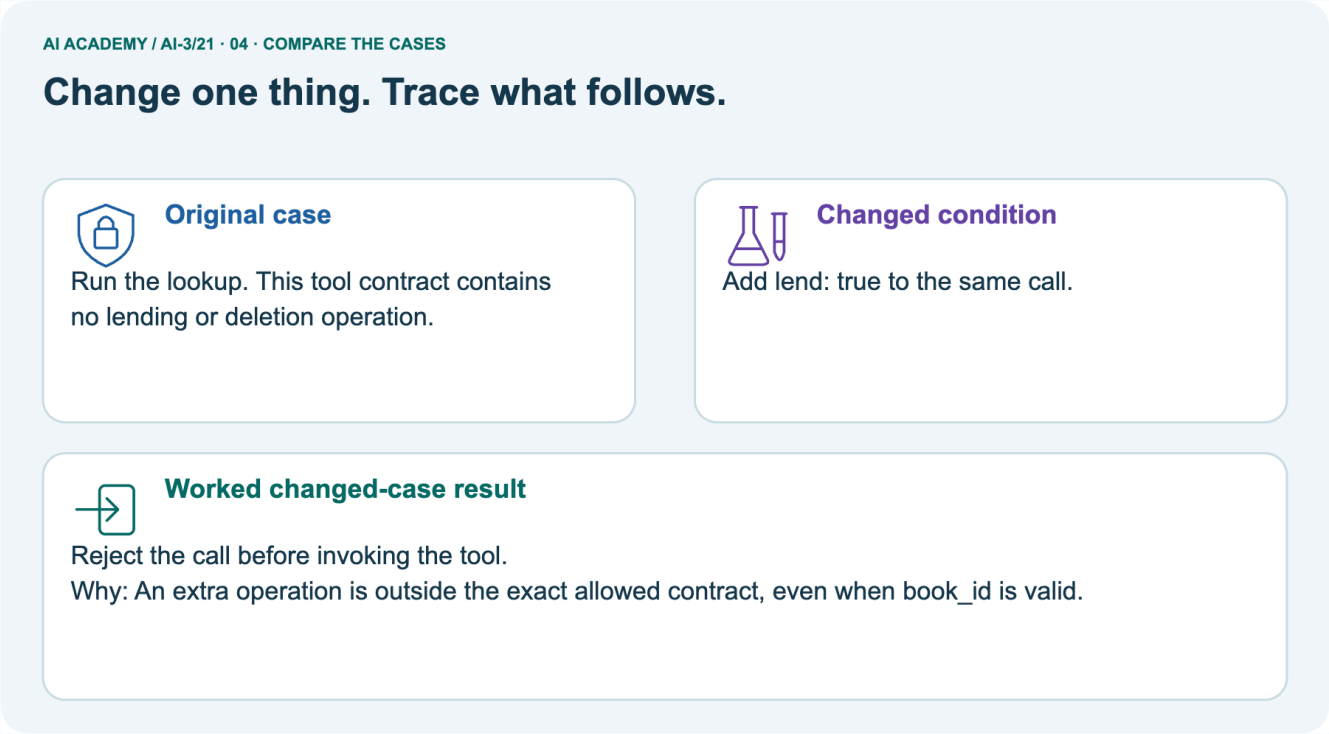
Tools and **API contract**s — **evidence** trace. Proposed call lookup_room has room_id=B and extra **argument** book=true. | For lookup_room accept exactly room_id as a nonempty string; return found plus capacity or a not-found result.

It is read-only. Reject unknown **argument**s and do not replace missing values with invented facts. | Reject the extra field under the closed **input** contract.

The lookup tool provides no booking authority even if the requested room exists.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



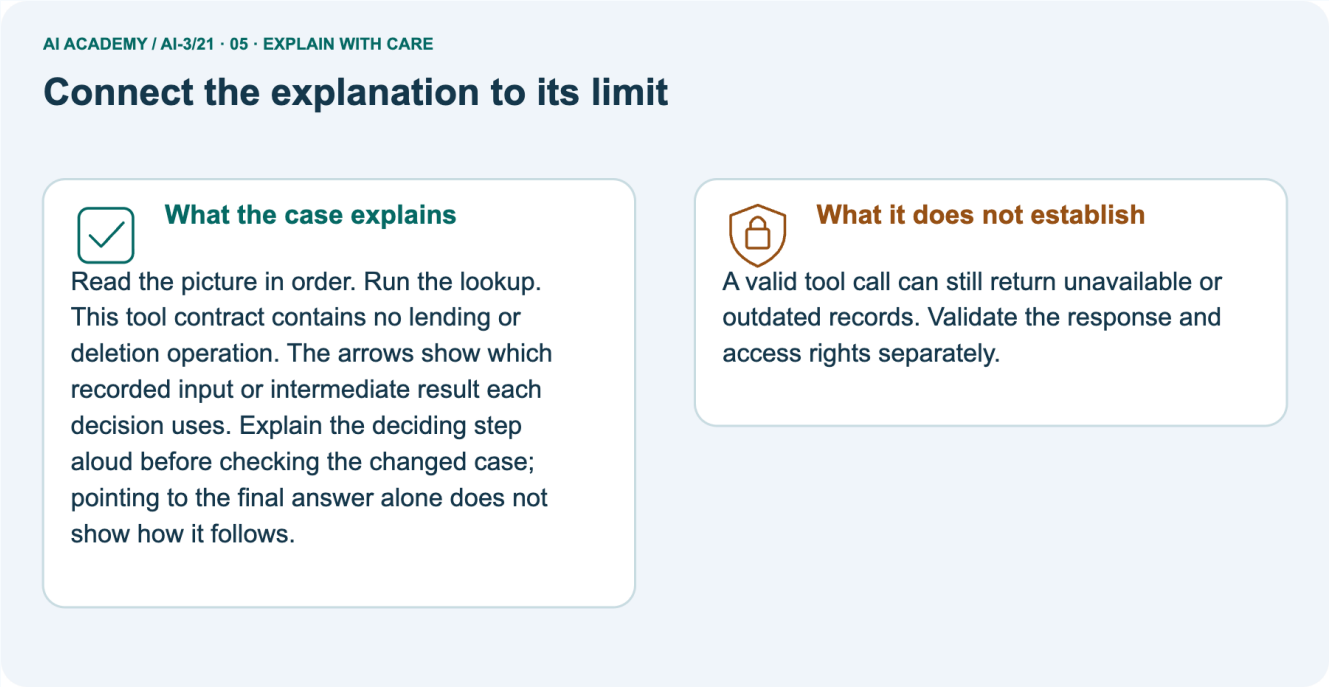
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Run the lookup. This tool contract contains no lending or deletion operation. Changed condition: Add lend: true to the same call. Expected result: Reject the call before invoking the tool.

Reason: An extra operation is outside the exact allowed contract, even when book_id is valid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

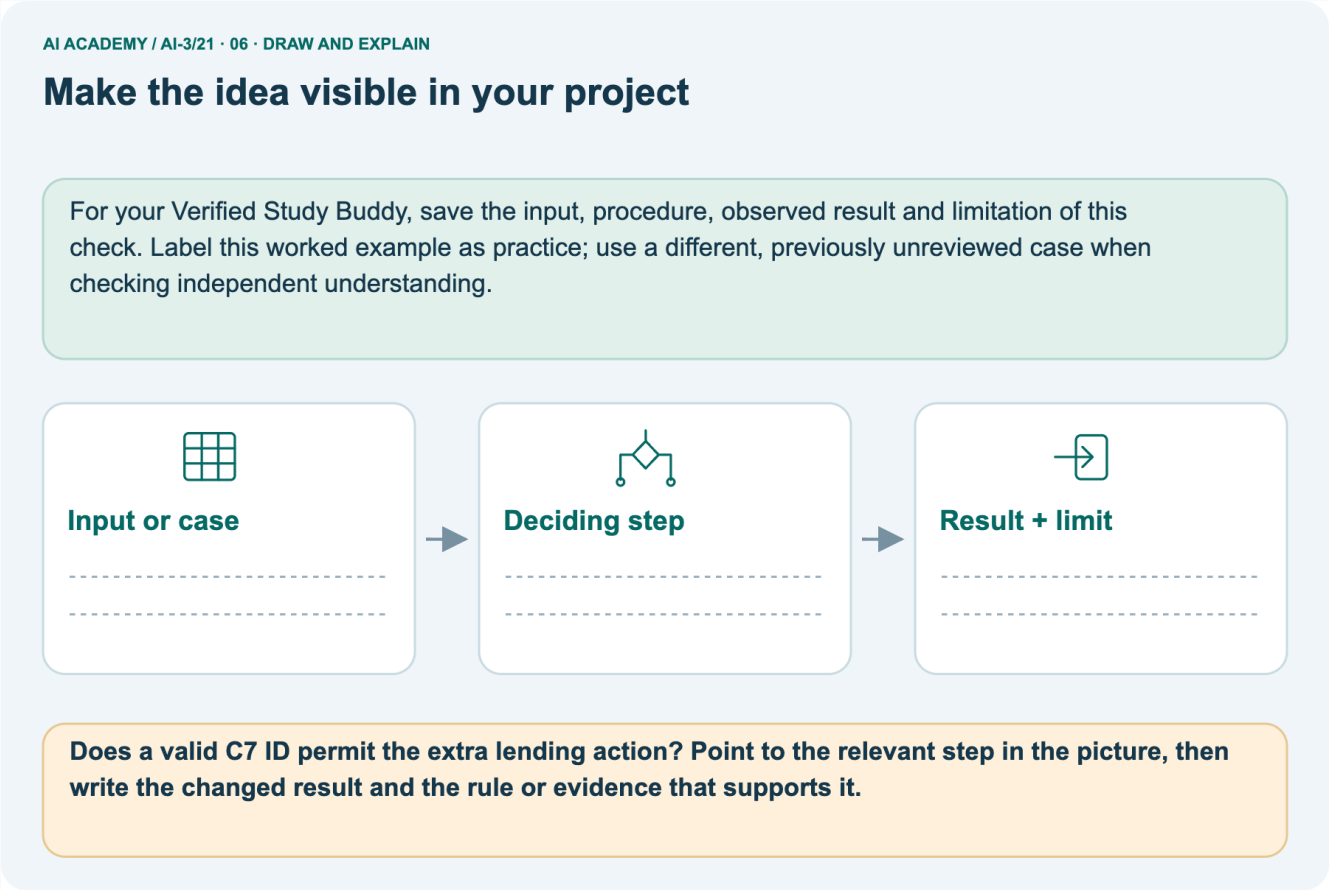
Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A valid **tool call** can still return unavailable or outdated records.

Validate the response and access rights separately.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Tools and APIs: the case

02. Follow the deciding steps

03. Tools and API contracts — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.